The First ChineseBabyLM Challenge: training data-efficient and cognitively plausible language models for Chinese

Siyuan Song[1*†], Zhiheng Qian[2†], Yunhao Zhang[3], Linyang He[4], Xiaozhe Ji[5], Yingxin Lin[6], Hongao Zhu[7], Chongtian Shao[2], Chuhan Lang[8], Luan Li[2], Rui Wang[2], Renfen Hu[5], Shaonan Wang[8], Hai Hu[8*]

[1]Princeton University. [2]Shanghai Jiao Tong University.
[3]Chinese Academy of Sciences. [4]Columbia University.
[5]Beijing Normal University. [6]Tsinghua University.
[7]University of California San Diego. [8]The Hong Kong Polytechnic University.

*Corresponding author(s). E-mail(s): ss1280@princeton.edu; chinese.babylm@gmail.com; hai.hu@polyu.edu.hk;
†These authors contributed equally to this work.

Abstract

This paper presents the first ChineseBabyLM Challenge, organized as part of NLPCC 2026. The challenge asked participants to train language models from scratch using no more than 102M Chinese words. The models were evaluated on three tracks: natural language understanding, cognitive alignment, and Hanzi knowledge. There were no restrictions on tokenizers, model architectures, or the number of training epochs. Eighteen teams submitted 28 distinct models, generating 74 result files. The overall-winning team used a DeBERTa-v2 architecture and introduced an auxiliary pinyin-prediction objective during pretraining. Several submissions also explored curriculum-learning strategies and architectural innovations. Overall, the challenge provides a benchmark for advancing data-efficient and cognitively plausible approaches to Chinese language modeling.

Keywords: BabyLM, Chinese, training, cognitively plausible

1 Introduction

Large language models have achieved strong performance across many language understanding tasks, but their success has typically depended on pretraining with very large corpora and substantial compute. This data-hungry paradigm contrasts sharply with human language acquisition: children acquire robust linguistic competence from comparatively limited, naturalistic input. The BabyLM Challenge was created to make this contrast experimentally useful by asking participants to train language models under developmentally motivated data budgets, encouraging research on sample-efficient pretraining, cognitively plausible data, and evaluation protocols that go beyond raw scale [1].

The original BabyLM shared tasks focused primarily on English. Recent multilingual efforts such as BabyBabelLM extend the same motivation to a wider set of languages, showing that data-efficient language modeling should be studied under typologically and orthographically diverse conditions [2]. Chinese is an especially important test case. It lacks explicit word boundaries, makes extensive use of compounding, permits flexible syntactic configurations, and uses a logographic writing system in which characters carry visual, structural, and phonological regularities. These properties make Chinese language learning a poor fit for evaluation protocols designed only around alphabetic, whitespace-delimited languages, and they raise distinct questions about tokenization, character-level representation, and the amount of data needed to acquire linguistic generalizations.

The ChineseBabyLM Challenge is a shared task for studying data-efficient and cognitively plausible language modeling for Chinese. Participating systems must be pretrained from randomly initialized weights, either on the official 102-million-word corpus derived from the Chinese portion of BabyBabelLM [2] or on a self-compiled corpus within the same word budget. This setup preserves the central BabyLM constraint while allowing participants to explore different model architectures, tokenizers, data selection strategies, and training methods for Chinese.

The challenge evaluates models along three complementary dimensions. The Natural Language Understanding track combines zero-shot minimal-pair evaluation, including ZhoBLiMP [3], with supervised Chinese understanding tasks from CLUE [4]. The Hanzi track directly tests whether models acquire structural and phonological knowledge about Chinese characters from limited input. The Cognitive Modeling track uses Chinese fMRI benchmarks from MulCogBench [5] to measure how well model representations predict human neural responses during word- and discourse-level language comprehension. Together, these tracks are intended to reward models that are not only accurate on downstream tasks, but also sensitive to Chinese-specific linguistic structure and aligned with cognitively motivated evaluation criteria.

This paper describes the design of the first ChineseBabyLM Challenge, including the training rules, evaluation tracks, task inventory, baseline systems, performance of the baseline models and models submitted by participating teams. By providing a standardized corpus, an open evaluation pipeline, hidden final-evaluation tasks, and reproducible baselines, the shared task aims to support a research community around Chinese data-efficient language modeling and to contribute a language-specific counterpart to the broader BabyLM program.

2 Evaluation Tasks

The ChineseBabyLM Challenge evaluates data-efficient Chinese language models along three complementary dimensions of linguistic competence: natural language understanding, character-level knowledge, and cognitive plausibility. The final evaluation consists of 14 tasks grouped into three tracks: eight Natural Language Understanding (NLU) tasks, four Hanzi tasks, and two Cognitive Modeling tasks.

Evaluation formats.

The tasks use three evaluation formats. First, the zero-shot minimal-pair tasks evaluate whether a model assigns a higher score to a well-formed or semantically plausible sentence than to a minimally different ill-formed or implausible one; the task score is the accuracy over all pairs. For the Hanzi tasks, if the target character or character components are not recognized by the tokenizer (UNK), then the model is considered to fail on that minimal pair.

Second, the CLUE-style NLU tasks are evaluated by task-specific fine-tuning for sequence classification or multiple-choice classification.

Third, the cognitive modeling tasks evaluate whether model representations can predict human fMRI responses: model hidden states are used as features in an encoding model, and the predicted neural responses are compared with observed fMRI responses using correlation-based metrics.

2.1 Natural Language Understanding Track

The NLU track measures syntactic and semantic understanding in Chinese. It contains two zero-shot minimal-pair tasks and six supervised fine-tuning tasks, all drawn from established Chinese benchmarks; Table 1 summarizes the inventory, splits, and sizes.

The zero-shot tasks test whether a pretrained model prefers the acceptable member of a minimal pair without task-specific training. The open task is ZhoBLiMP [3], whose 118 paradigms (35,400 pairs in total) cover a broad range of Chinese syntactic and semantic phenomena. The hidden task is XCOMPS-ZH [6], 14,368 pairs targeting concept–property compatibility, which complements the primarily morphosyntactic focus of ZhoBLiMP with commonsense-like semantic judgments.

The six supervised tasks are evaluated by task-specific fine-tuning. The open tasks are AFQMC (sentence-pair semantic matching), TNEWS (short-text news classification), and CLUEWSC2020 (Winograd-style coreference) from CLUE [4], together with OCNLI (natural language inference) [7]. The hidden tasks are C3, a multiple-choice reading comprehension benchmark [8], and a diagnostic NLI task [9]; both provide held-out evaluation beyond the open development tasks.

2.2 Cognitive Modeling Track

The Cognitive Modeling track evaluates whether model representations align with human brain responses during Chinese language comprehension. The two tasks are derived from the Chinese fMRI components of MulCogBench [5]. In both tasks, the evaluation pipeline extracts model representations and fits an encoding model from

Track	Task	Split	Size in final files	Target ability
NLU	ZhoBLiMP	Open	35,400 minimal pairs, from 118 paradigms	Zero-shot grammatical and semantic judgments
NLU	XCOMPS-ZH	Hidden	14,368 minimal pairs	Zero-shot concept–property compatibility judgments
NLU	AFQMC	Open	34,334 train / 4,316 dev records	Sentence-pair semantic similarity
NLU	OCNLI	Open	50,437 train / 2,950 dev records	Natural language inference
NLU	TNEWS	Open	53,360 train / 10,000 dev records	Short-text news classification from news titles
NLU	CLUEWSC2020	Open	1,244 train / 304 dev records	Chinese Winograd-style coreference resolution
NLU	C3	Hidden	11,869 train / 3,816 dev records	Multiple-choice Chinese machine reading comprehension
NLU	diagnostic_nli	Hidden	50,437 train / 2,122 dev records	Diagnostic natural language inference
Hanzi	`hanzi_structure`	Open	2,000 minimal pairs	Character-level structural knowledge
Hanzi	`hanzi_structure_hidden`	Hidden	2,000 minimal pairs	Held-out character-structure minimal pairs
Hanzi	`hanzi_pinyin`	Open	2,000 minimal pairs	Character-level phonological knowledge
Hanzi	`hanzi_pinyin_hidden`	Hidden	2,000 minimal pairs	Held-out character-phonology minimal pairs
Cog	`word_fmri`	Open/Hidden	672 word stimuli in the source fMRI dataset	Brain-aligned word-level evaluation: fit model representations to fMRI responses
Cog	`fmri`	Open/Hidden	60 story stimuli, 52,269 words in the source fMRI dataset	Brain-aligned discourse-level evaluation: fit model representations to fMRI responses

Table 1 Task inventory for the final ChineseBabyLM evaluation. Sizes for NLU and Hanzi tasks are from the final evaluation files; sizes for the Cog tasks describe the corresponding Chinese fMRI stimuli from MulCogBench.

those representations to fMRI responses. Better prediction performance indicates that the semantic information encoded in the model's representations is more similar to that encoded in the brain's neural representations elicited by the corresponding linguistic stimuli [10].

The `word_fmri` task is a word-level brain-alignment benchmark based on a Chinese word fMRI dataset introduced by Wang et al. [11]. The dataset contains neural responses from 11 participants as they read 672 isolated Chinese words. The task evaluates whether representations of individual words encode information predictive of the spatial patterns of fMRI responses elicited during word comprehension. This task therefore probes relatively local lexical and semantic representations.

The `discourse_fmri` task is a discourse-level brain-alignment benchmark based on a Chinese discourse fMRI dataset introduced by Wang et al. [12]. The dataset contains neural responses from 12 native Chinese speakers as they listened to 60 stories comprising 52,269 words in total. Compared with the word-level task, this

task evaluates representations in a more natural and context-rich comprehension setting. It tests whether models encode information useful for predicting neural activity during extended spoken discourse, where lexical, syntactic, semantic, and discourse-contextual information must be integrated over time.

2.3 Hanzi Track

The Hanzi track evaluates knowledge of properties specific to the Chinese writing system, namely Hanzi structure and Hanzi pinyin (see Table 2). Unlike the CLUE tasks, which focus on sentence- or passage-level understanding, the Hanzi tasks test whether a model has learned character-level regularities from limited pretraining data. All four Hanzi tasks are evaluated as zero-shot minimal-pair tasks. Each file contains 2,000 minimal pairs.

For the two Hanzi structure tasks, target characters are drawn from the Level-1 list of the *General Standard Chinese Character Table* [13]. Character structures and component decompositions follow the Chinese Character Dataset of Wu et al. [14]. Phonetic-role and IDS-based annotations use `ids-analysis.txt` from CJKVI-IDS [15]. Target character components must appear at least 50 times in the official pretraining corpus.

For Hanzi pinyin tasks, target characters are restricted to single-pronunciation (monophonic) characters drawn from the 3,500 most frequent Chinese characters, so that every item has an unambiguous reading. Pinyin readings are obtained with `pypinyin` (manually corrected where mislabeled) and mapped to IPA with `Dragonmapper`, and each syllable is decomposed into three independent dimensions: the *initial* (onset consonant), the *final* (rime), and the lexical *tone* (1–4). All pinyin items take the form of homophone judgments: each minimal pair contrasts a true statement that two characters share an identical reading with a minimally different false statement in which the second character is replaced by one that differs in exactly one of the three dimensions, yielding the `initial_diff`, `rhyme_diff`, and `tone_diff` conditions. There is also a `disjoint` condition, where the two characters have completely different initials, rhymes and tones. To ensure that the judgment reflects phonological knowledge rather than lexical frequency, the two characters in each pair are matched on corpus frequency via Zipf scores.

3 Rules for model training

All participating groups are required to pretrain their models from scratch (that is, randomly initialized weights). Loading pretrained checkpoints or distilling from existing large models is not allowed.

There are two options for pretraining data. Option 1 is to use the official corpus released by the organizers, a 102M-word training corpus sampled from the BabyBabelLM project [2][1]. Option 2 is to use a self-compiled pretraining corpus, which should contain at most 102M words counted via Jieba 0.42.1 in default mode [2]. There is no restriction on the number of training epochs.

[1] https://huggingface.co/datasets/chinese-babylm-org/babylm-zho-100M
[2] https://github.com/fxsjy/jieba

Task	Subtype	Good example	Bad example
Structure	whole-to-component	阁字外围的部分是门。	阁字外围的部分是包。
Structure	component-to-whole	外围是门，内部是圭，这个字是闺。	外围是门，内部是圭，这个字是盔。
Pinyin	`initial_diff`	“交”和“郊”的声母、韵母和声调完全相同。(jiao1, jiao1)	“交”和“销”的声母、韵母和声调完全相同。(jiao1, xiao1)
Pinyin	`rhyme_diff`	“机”和“鸡”的声母、韵母和声调完全相同。(ji1, ji1)	“机”和“兢”的声母、韵母和声调完全相同。(ji1, jing1)
Pinyin	`tone_diff`	“欲”和“芋”的声母、韵母和声调完全相同。(yu4, yu4)	“欲”和“于”的声母、韵母和声调完全相同。(yu4, yu2)
Pinyin	`disjoint`	“纠”和“鸠”的声母、韵母和声调完全相同。(jiu1, jiu1)	“纠”和“缠”的声母、韵母和声调完全相同。(jiu1, chan2)

Table 2 Representative minimal-pair examples in the Hanzi track. Structure items test character composition; pinyin items are homophone judgments built on the template “X and Y have identical initial, final, and tone”. The good example is a true statement (Y is a homophone of X); the bad example replaces Y with a distractor that differs in exactly one dimension (`initial_diff`, `rhyme_diff`, `tone_diff`) or in all three (`disjoint`). Tone-numbered pinyin is shown in parentheses.

We do not have any special requirements on the tokenizer or model architecture. Note that the two Hanzi tasks are designed specifically to encourage innovations on tokenizer design so that the models encode natively the phonological and orthographical information of Chinese characters.

4 Baseline models and results

We adopted six models as baselines for the first ChineseBabyLM challenge. In addition to four models pretrained from scratch, we evaluated Qwen3-0.6B[3] and BERT-base-Chinese[4] as two established open-source models trained on large-scale corpora. The implementation details of the baselines are as follows.

For tokenizers, we randomly sampled roughly 0.8B characters from CCI3-HQ [16], a high-quality Chinese pretraining corpus, and trained a byte-level BPE tokenizer with a 50,000 vocabulary size and a minimum frequency threshold of 2. The tokenizer was used across all the baseline models we trained from scratch.

In terms of model architecture, we trained four architecture-diverse baseline models: an encoder-decoder Transformer following T5 [17], a bidirectional encoder following BERT [18], a selective state-space model following Mamba [19], and a decoder-only Transformer following Pythia [20]. The decoder-only Pythia and Mamba baselines are optimized with causal language modeling, while the BERT baseline uses masked language modeling with a masking probability of 15%. The T5 baseline is trained as a text-to-text continuation model: each training sequence is split into a source prefix and target continuation, and the model is optimized to generate the continuation from the prefix. To make the comparison computationally controlled, all architectures are scaled down to approximately 14M parameters and are trained from scratch on the official ChineseBabyLM training corpus.

The baseline models were trained for 20 epochs using the Adam optimizer, with a learning rate of 5×10^{-4}, a batch size of 32, and a 1% warmup ratio. The context

[3] https://huggingface.co/Qwen/Qwen3-0.6B
[4] https://huggingface.co/google-bert/bert-base-chinese

length was set to 1024. The 3-, 10-, and 20-epoch checkpoints were released to the candidates.

Architecture	NLU Track	Hanzi Track	Cog Track
T5	54.17	50.71	31.63
BERT	<u>57.65</u>	51.86	<u>33.38</u>
Mamba	56.89	<u>52.49</u>	28.28
Pythia	55.88	51.73	29.28
Qwen3-0.6B	57.82	**61.24**	**34.85**
BERT-base-Chinese	**66.00**	57.13	33.99

Table 3 Average performance of the baseline models by track. Bold indicates the best result among all evaluated models, while underlining indicates the best result among the four 14M models pretrained from scratch.

Table 3 presents the average performance of all baseline models across three tracks. Among the four 14M models pretrained from scratch, BERT achieves the highest average scores on the NLU and Cognitive Modeling tracks, whereas Mamba performs best on the Hanzi track. Among the two external reference models, BERT-base-Chinese obtains the highest NLU score, while Qwen3-0.6B leads on the Hanzi and Cognitive Modeling tracks. These two models do not satisfy the ChineseBabyLM data constraint and are only reported as external reference points.

5 Shared Task

5.1 Overview

The final stage of the first ChineseBabyLM Challenge received submissions from 18 teams. In total, the teams submitted 28 distinct models, with several teams entering more than one model or rerunning evaluations during the submission period. The evaluation records contain 74 result files across these initial submissions and reruns. For the official team-level ranking, one system from each team was retained, resulting in 18 systems on the final leaderboard.

For every ranked system, task scores are reported on a 0–100 scale. Each track score is the unweighted mean of its constituent tasks, and the overall score is the macro-average of the NLU, Cognitive Modeling, and Hanzi track scores. Missing task results are assigned a score of zero. This track-level macro-average prevents tracks with more constituent benchmarks from receiving greater weight solely because of their size.

5.2 Shared task results

Table 4 reports the official team-deduplicated final leaderboard. The best award-eligible score in each aggregate column is shown in bold.

According to the submitted system metadata and technical report, BERT is the most common architecture, accounting for 11 of the 18 ranked systems. Five systems

Rank	Team	Model	Arch.	NLU	Cog	Hanzi	Overall
1	BabyDragon	BabyDragon-v4	DeBERTa-v2	59.98	32.94	**91.56**	**61.49**
2	OK的老大	babydeberta-24m-hanzi-v8s	BERT	58.55	33.35	81.46	57.79
3	TUM Campus Heilbronn	babylm-zho-pinyin-code-97M	GPT-2	56.40	32.80	75.51	54.90
4	SMI	Chinese_babylm	BERT	60.13	33.78	61.41	51.77
5	lampa	chinese-babylm-2026-hanzi-a3large	GPT-2	59.09	33.32	58.70	50.37
6	YNU-HPCC	BabyLM-Chinese-DeBERTa-V2-14M	DeBERTa-v2	**60.99**	33.29	55.71	50.00
7	西湖往事	babylm-zh-run10a	GPT-2	60.75	33.27	54.69	49.57
8	L-Cluster	chinese-babylm-2026-v3	BERT	58.65	33.11	56.49	49.42
9	moddells	moddells/babylm	BERT	58.99	33.67	52.70	48.45
10	CogTeam	babyconceptLM-baike102M	BERT	56.79	33.76	53.61	48.05
11	Alice	Chinese BabyLM NanoChat D22	GPT-2	56.87	33.97[5]	50.38	47.07
12	Flops	pinyinboost-bert-10M-6L384H-stage6e-vocabadapt	BERT	52.34	32.16	51.59	45.36
13	KABUTOMUSHI	chinese-babylm-bpe-11	BERT	53.21	30.38	49.96	44.52
14	所以然	syr_model_bert_final_3	BERT	50.14	30.91	50.10	43.72
15	XJTLU-SeedLM	GPT2-ChineseDevBench	GPT-2	43.98	34.10[6]	27.67	35.25
16	ZZUnlp	public_my_babys10	BERT	59.02	33.35	1.19	31.19
17	Deepscientist	chinese-babylm-cog-a197-strict-best	BERT	0.00	**33.96**	0.00	11.32
18	kiikii	neurobert-cog-v2	BERT	0.00	33.00	0.00	11.00

[5]This team's submitted result used the best layer.

[6]This submission was evaluated only on open tasks and data and was therefore ineligible for an award.

Table 4 Official results of the ChineseBabyLM shared task. NLU, Cog, and Hanzi are the mean scores for the three evaluation tracks; Overall is their macro-average. The best score in each aggregate column is shown in bold.

use GPT-2 and two use DeBERTa-v2; grouping BERT and DeBERTa-v2 together, BERT-family architectures therefore account for 13 of the 18 entries.

The two highest-ranked systems are encoder-only, whereas the third-ranked system is decoder-only. All three track-winning systems are encoder-only, although the strong overall performance of the third-ranked decoder-only system shows that competitive submissions were not limited to a single model type.

The top submissions to the first ChineseBabyLM challenge illustrate three complementary approaches to learning Chinese under a limited-data regime: writing-system-aware pretraining, brain-aligned representation synthesis, and compact curriculum-trained language modeling.

BabyDragon-v4, submitted by the BabyDragon team, ranked first overall and first on the Hanzi track [21]. It is a 104M-parameter DeBERTa-v2 encoder trained from scratch for 60 epochs with whole-word masked language modeling and an auxiliary pinyin-prediction objective. Its 92.91M-word training mixture contains 6.76M words of injected facts. Pinyin and character structure facts were derived from resources such as `pypinyin`, CHISE/CJKVI-IDS, and Unihan, while grammar examples were generated by rule; no large language model was used to generate or paraphrase the data. Controlled analysis indicates that the model generalizes well to homophone pairs not explicitly injected, whereas its orthographic decomposition performance depends much more strongly on coverage of the injected character facts. Although the tokenizer

was extended with 329 radical and component tokens, this extension was effectively inactive on the final evaluation because all tested components were already covered.

chinese-babylm-cog-a197-strict-best (RepDuet) [22] won the Cognitive Modeling track after organizer reproduction and final-layer verification. The two higher raw Cog entries were excluded from the award because one used only open tasks and data and the other did not submit final-layer representations. RepDuet uses two compact BERT encoders trained from scratch on the official corpus and constructs a different representation for each fMRI task. The word-level representation concatenates intermediate contextual states with random-indexing and token frequency features. The discourse-level representation combines an upper-layer state mixture with lexical statistics and MLM-derived confidence features, including token likelihood, entropy, maximum probability, and prediction margin. This task-specific design achieved a verified Cog score of 33.96.

BabyLM-Chinese-DeBERTa-V2-14M (BabyDeBERTa-zh) [23] won the NLU track with a score of 60.99. It is a 14M-parameter DeBERTa-v2 masked language model with 12 layers, a hidden size of 256, a feed-forward size of 1024, and 8 attention heads. Its 12,800-token WordPiece tokenizer was retrained on the official corpus. The main innovation is a deterministic four-stage curriculum that restricts complete training sequences to maximum lengths of 32, 64, and 128 before exposing the model to the full-length corpus. In the final 30-epoch run, the four stages receive 2, 3, 5, and 20 epochs, respectively. Compared with an otherwise comparable non-curriculum baseline, this schedule raises the NLU average from 59.70 to 60.99, with the largest improvement on C3.

babydeberta-24m-hanzi-v8s (KADeBERTa-v2) [24] ranked second overall and second on the Hanzi track. It is a compact DeBERTa-v2 encoder with approximately 26M parameters, 10 layers, a hidden size of 384, and 6 attention heads. Its custom 12,119-token WordPiece vocabulary preserves Chinese character components as atomic units. The model augments the official corpus with automatically generated pinyin and character-structure statements and uses weighted sampling to expose the model more frequently to this small auxiliary corpus during masked-language-model pretraining. Ablation results show that phonological facts primarily improve pinyin judgments, whereas structural facts primarily improve character-structure judgments.

babylm-zho-pinyin-code-97M, submitted by TUM Campus Heilbronn, ranked third overall [25]. The system replaces Hanzi input with an *Initial+Digit* representation that maps each pinyin syllable to two ASCII symbols encoding its initial and coarse information about tone and syllable length. Codes within each Jieba-segmented word are concatenated, while word boundaries are retained; the same deterministic conversion is applied to evaluation inputs. A 16,000-token SentencePiece BPE tokenizer is trained on this representation and used with a 97.7M-parameter, 12-layer decoder-only GPT model optimized by causal language modeling. Its results show that a heavily compressed phonetic representation can retain enough information for competitive Chinese language modeling despite discarding Hanzi identity and most orthographic information.

6 Discussion

6.1 Approaches of the top systems

The five award-winning systems adopt complementary strategies. The two highest-ranked Hanzi systems, BabyDragon-v4 and KADeBERTa-v2, both improve a standard DeBERTa-v2 encoder primarily through *data augmentation*, injecting rule-generated pinyin and character-structure facts into pretraining; their success confirms that the character-level knowledge probed by the Hanzi track is largely absent from ordinary child-directed text and must be supplied explicitly. The NLU winner, BabyDeBERTa-zh, takes a different route, relying on a sequence-length curriculum rather than augmented data. The Cognitive Modeling winner, RepDuet, changes neither the model nor the data but the *representations used at evaluation*, combining several compact BERT checkpoints into task-specific feature sets. The third-ranked system, babylm-zho-pinyin-code-97M, is the only top system to innovate on the *tokenizer and input representation* itself, replacing Hanzi with a compressed phonetic code that retains enough information for competitive language modeling while discarding orthographic identity.

This contrast points to several directions for future modeling. Because the Hanzi winners relied on data augmentation, an open question is whether comparable gains can be obtained through architectural or objective-level inductive biases for phonology and orthography rather than injected facts. More broadly, babylm-zho-pinyin-code-97M shows that representation design is a promising but underexplored axis: future systems could explore tokenizers that encode orthographic, phonological, and semantic information natively, potentially without external augmentation. Finally, as each award-winning system was optimized for a single track, building one model that is competitive across NLU, Hanzi, and cognitive modeling remains an open goal.

6.2 Task discriminability

The three tracks differ markedly in how well they separate systems. Across the 18 ranked submissions, the Hanzi track spans roughly 90 points (1.19–91.56) and the NLU track 17 points (43.98–60.99 among the teams that submitted it), whereas the Cognitive Modeling track spans fewer than 4 points (30.38–34.10, standard deviation below 1). The Cognitive Modeling compression is driven by its constituent tasks: the word-level fMRI task varies by only 1.2 points over the entire field (55.47–56.65), and the discourse-level task, although somewhat wider (5.17–11.90), does so at low absolute scores. In its current form the track therefore contributes little to the overall ranking once a system produces reasonable representations.

Several NLU tasks are similarly weak at discriminating systems. AFQMC varies by only 2.3 points across all submitting teams (68.86–71.11), with most systems within a single point of one another—the same label-bias degeneracy observed in the baselines, now seen across the whole field, indicating that this task measures little about the models. CLUEWSC2020 is also compressed (62.83–68.75). By contrast, the minimal-pair and inference tasks discriminate well: ZhoBLiMP spans 29 points (53.21–82.21), OCNLI 19 (53.56–72.68), and the diagnostic NLI task 16 (38.31–54.48). The Hanzi

tasks separate the field strongly but exhibit ceiling effects among the top systems, with two teams reaching 99.85 and 95.80 on open pinyin.

These observations motivate a revision of the task inventory in future editions. Tasks that fail to discriminate among systems—*word_fmri*, AFQMC, and CLUEWSC2020 being the clearest cases—should be revised or retired, while high-spread tasks such as ZhoBLiMP, the NLI tasks, and the Hanzi tasks should be retained and, where they saturate, made harder. Adding new tasks that target underrepresented Chinese phenomena, such as word segmentation, morphology, or discourse coherence, or developmental tasks such as Age of Acquisition prediction, would further increase the diagnostic value of the benchmark.

6.3 Comparison with the English BabyLM Challenge

Comparing the first ChineseBabyLM Challenge with the English BabyLM shared tasks highlights both shared and language-specific design choices.

Model architectures.

The dominant architectures differ in instructive ways. In the English BabyLM challenges, decoder-only causal models and hybrid objectives were prominent among the leading systems; in particular, GPT-BERT, which combines causal and masked language modeling in a single stack, was a winning entry [26]. In ChineseBabyLM, by contrast, no team adopted such a causal/bidirectional hybrid: encoder-only BERT and DeBERTa-v2 systems dominated, accounting for 13 of the 18 ranked entries. This divergence is consistent with a controlled finding from the top-ranked BabyDragon team, who reported that adding a GPT-BERT-style causal branch was the weakest of their architectural variants, lowering rather than raising performance [21]. They attribute this to the all-bidirectional nature of the ChineseBabyLM evaluation, in which a causal signal dilutes the masked objective without an evaluation payoff. The contrast suggests that the value of a modeling innovation depends on the evaluation regime, and that recipes validated on English may not transfer automatically to Chinese.

Task selection.

The task inventories overlap but diverge where the languages do. Both challenges include a natural-language-understanding track built on grammatical minimal pairs (BLiMP for English, ZhoBLiMP for Chinese) and supervised classification, and both evaluate cognitive plausibility. Beyond this shared core, the English challenge additionally includes behavioral psycholinguistic tasks such as age-of-acquisition prediction and reading-time modeling [27–29], which ChineseBabyLM does not yet have. Conversely, the Hanzi track is specific to Chinese, probing orthographic structure and phonology that have no direct English counterpart. Future ChineseBabyLM editions could therefore both expand the shared cognitive component—for example by adding Chinese reading-time or acquisition-trajectory tasks to parallel the English setup—and deepen the Chinese-specific component that currently distinguishes the two benchmarks.

7 Conclusion and future directions

In this paper, we have described the design and results of the first ChineseBabyLM Challenge, which was created to promote data-efficient and cognitively plausible language modeling for Chinese. The submitted systems illustrate several complementary strategies for learning under a limited-data regime: compact DeBERTa-style encoders; Chinese-specific tokenization, pinyin-aware objectives, and explicit character-knowledge augmentation; curriculum learning; compressed phonetic input representations; and task-specific combinations of contextual, lexical, and uncertainty features for cognitive modeling.

These findings suggest several directions for future ChineseBabyLM research. First, systematic ablations would help clarify the contribution of knowledge injection, tokenizer design, pinyin-aware objectives, curriculum schedules, phonetic recoding, and multi-checkpoint representation fusion. Second, future challenges could encourage unified models that perform well across NLU, Hanzi, and cognitive modeling rather than optimizing for a single track. Finally, Chinese offers rich opportunities for language-specific evaluation beyond standard sentence understanding, including orthographic structure, phonology, morphology, reading difficulty, and developmental acquisition. Incorporating these factors into both pretraining and evaluation may produce models that are not only more accurate but also more informative as computational models of Chinese language learning.

References

[1] Warstadt, A., *et al.*: Findings of the BabyLM challenge: Sample-efficient pretraining on developmentally plausible corpora. In: BabyLM@CoNLL 2023, pp. 1–34 (2023)

[2] Jumelet, J., *et al.*: Babybabellm: A multilingual benchmark of developmentally plausible training data. In: EACL, pp. 3297–3329 (2026)

[3] Liu, Y., *et al.*: A systematic assessment of language models with linguistic minimal pairs in Chinese. TACL **14**, 755–771 (2026)

[4] Xu, L., *et al.*: CLUE: A Chinese language understanding evaluation benchmark. In: COLING, pp. 4762–4772 (2020)

[5] Zhang, Y., *et al.*: Mulcogbench: a multi-modal cognitive benchmark dataset for evaluating Chinese and English computational language models. Lang. Resour. Eval. **59**(3), 3005–3028 (2025)

[6] He, L., *et al.*: XCOMPS: A multilingual benchmark of conceptual minimal pairs. In: SIGTYP 2025, pp. 75–81 (2025)

[7] Hu, H., *et al.*: OCNLI: Original Chinese Natural Language Inference. In: Findings EMNLP, pp. 3512–3526 (2020)

[8] Sun, K., Yu, D., Yu, D., Cardie, C.: Investigating prior knowledge for challenging Chinese machine reading comprehension. TACL **8**, 141–155 (2020)

[9] Hu, H., *et al.*: Investigating transfer learning in multilingual pre-trained language models through Chinese natural language inference. In: Findings ACL, pp. 3770–3785 (2021)

[10] Zhang, Y., *et al.*: Navigating brain language representations: A comparative analysis of neural language models and psychologically plausible models. In: CogSci 2024, vol. 46 (2024)

[11] Wang, S., *et al.*: An fmri dataset for concept representation with semantic feature annotations. Sci. Data **9**(1), 721 (2022)

[12] Wang, S., Zhang, X., Zhang, J., Zong, C.: A synchronized multimodal neuroimaging dataset for studying brain language processing. Sci. Data **9**(1), 590 (2022)
[13] Ministry of Education and State Language Commission of China: Table of General Standard Chinese Characters. Level-1 list: 3,500 characters (2013). http://www.gov.cn/gzdt/att/att/site1/20130819/tygfhzb.pdf
[14] Wu, X., Stratos, K., Xu, W.: The impact of visual information in Chinese characters: Evaluating large models' ability to recognize and utilize radicals. In: NAACL-HLT 2025, pp. 331–350 (2025). https://aclanthology.org/2025.naacl-long.16/
[15] Kawabata, K., Nakamura, M., Huang, J.: CJKVI-IDS: Ideographic Description Sequences for CJK Unified Ideographs (2018). https://github.com/cjkvi/cjkvi-ids
[16] Wang, L., et al.: CCI3.0-HQ: a large-scale Chinese dataset of high quality designed for pre-training large language models (2024)
[17] Raffel, C., *et al.*: Exploring the limits of transfer learning with a unified text-to-text transformer. JMLR **21**(140), 1–67 (2020)
[18] Devlin, J., Chang, M.-W., Lee, K., Toutanova, K.: Bert: Pre-training of deep bidirectional transformers for language understanding. In: NAACL-HLT 2019, pp. 4171–4186 (2019)
[19] Gu, A., Dao, T.: Mamba: Linear-time sequence modeling with selective state spaces (2023)
[20] Biderman, S., *et al.*: Pythia: A suite for analyzing large language models across training and scaling. In: ICML, pp. 2397–2430 (2023)
[21] Zhao, Y., Zhang, Y., Chen, S., Liu, Y., Han, Z.: What a baby language model generalizes and what it must memorize: The winning system for the NLPCC 2026 Chinese BabyLM challenge. In: NLPCC 2026 Shared Task (2026)
[22] Liu, Z., Xie, Q., Zhu, M., Ning, Z., Weng, Y., *et al.*: RepDuet at NLPCC 2026 Chinese BabyLM: Task-specific hybrid representations for the Cognitive track. In: NLPCC 2026 Shared Task (2026)
[23] Lu, R., Wang, J., He, K., Xu, D., Zhang, X.: Curriculum-enhanced BabyDeBERTa for natural language understanding in Chinese. In: NLPCC 2026 Shared Task (2026)
[24] Cai, J., Deng, X., Chen, Y., Xu, K.: Beyond natural text: Teaching a BabyLM Chinese character form and sound. In: NLPCC 2026 Shared Task (2026)
[25] Robrecht, T., Xie, P., Sun, M., Gebert, J., Edman, L., *et al.*: Compressed Pinyin-based representation for Mandarin language modeling. In: NLPCC 2026 Shared Task (2026)
[26] Charpentier, L.G.G., Samuel, D.: Gpt or BERT: why not both? In: Proceedings of the 2nd BabyLM Challenge Workshop (2024). https://aclanthology.org/2024.conll-babylm.24/
[27] Warstadt, A., *et al.*: Findings of the BabyLM challenge: Sample-efficient pretraining on developmentally plausible corpora. In: BabyLM@CoNLL 2023, pp. 1–34 (2023)
[28] Hu, M.Y., *et al.*: Findings of the second BabyLM challenge: Sample-efficient pretraining on developmentally plausible corpora. In: BabyLM@CoNLL 2024, pp. 1–21 (2024)
[29] Charpentier, L., *et al.*: Findings of the third BabyLM challenge: Accelerating language modeling research with cognitively plausible data. In: BabyLM Workshop 2025, pp. 399–420 (2025)